Naveen Rajan

Machine Learning Engineer · computer vision & autonomous systems

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PROFILE

Software and ML engineer with five years of experience building AI systems integrated with real-world software infrastructure. Strong in system architecture and end-to-end ownership, with a track record of taking solutions from 0 to 1. Currently extending that background into LLM-based systems, agentic workflows and knowledge graphs.

Available immediately. Full-time, working-student, internship and freelance/contract roles in ML, computer vision and software engineering. Vienna on-site or remote within the EU.

EXPERIENCE

Founding Engineer, Computer Vision & Autonomy, Flying4Home (clearBee) 2021 – 2026

Early engineer and co-founder of an autonomous drone startup (5–8 people). Progressed from part-time prototyping while studying to full-time technical ownership of the perception and navigation stack.

- Improved successful A→B autonomous indoor flights from <30% to ~97% in arbitrary indoor environments via multimodal recovery policies and sensor fusion (vision + IMU + fiducial).
- Achieved a stable 20 Hz control loop and inference on edge Android hardware.
- Architected a hierarchical Kotlin coroutine orchestration system for concurrent perception, telemetry and control; eliminated lifecycle leaks, zombie states and cross-pipeline contention.
- Combined classical CV (OpenCV, IMU/Kalman filtering) with pretrained monocular depth models (Depth Anything, MiDaS, ZoeDepth).
- Built system-wide observability with Grafana, Prometheus and Loki to diagnose intermittent hardware and software failures; experiment tracking with ClearML.
- Designed the full frontend, backend and database for the multi-platform control app (Android, iOS, React/FastAPI web) with Firebase auth and realtime database.

Full-Stack Developer, Self-employed 2018 – 2021

Built and deployed a localized e-commerce service for a niche grocery market during COVID-19 (Python backend, REST APIs, React frontend, Firebase), owning it from architecture through deployment, integrations and user support.

SELECTED PROJECTS

LLM entity resolution · github.com/NavRjn/entity-resolution 2025 – present

Documents to knowledge graphs: Docling parsing, gpt-4o extraction into a strict Pydantic ontology, FastAPI + React review UI, Langfuse tracing, ground-truth evaluation (0.71 F1, work in progress), offline test suite.

PINN research framework · github.com/NavRjn/PINN-Thesis 2025 – present

Modular PyTorch/Typer CLI framework with runtime problem plugins and YAML-reproducible experiments; Vision Transformers and attention-based conditioning for PDE families.

EDUCATION

BSc Artificial Intelligence, Johannes Kepler University (JKU) Oct 2023 – expected early 2027

Thesis on physics-informed neural networks: conditioning latent-space manifolds, comparing attention-based mechanisms with vector-set representations to solve under-determined families of nonlinear PDEs with a single model. Mathematics and statistics average 1.6; NLP practical grade 1 (Sehr gut).

SKILLS

Languages: Python, Kotlin, TypeScript, C++, SQL, Bash, Java, C#

ML & LLM systems: PyTorch, ONNX, Hugging Face, LLMs & RAG, LangChain, Langfuse, Ollama, OpenAI API, Vector databases, PINNs, Vision Transformers

Computer vision & perception: OpenCV, YOLO, Monocular depth (Depth Anything, MiDaS, ZoeDepth), Sensor fusion, IMU / Kalman filtering, Tracking & segmentation

Backend & data: FastAPI, Pydantic, REST, PostgreSQL, MongoDB, Firebase, Async pipelines

Frontend & mobile: React, Astro, Android (Jetpack Compose, Coroutines), iOS integration

Infra & observability: Docker, CI/CD, Git/GitHub/GitLab, Grafana, Prometheus, Loki, ClearML, Linux/WSL

Robotics & embedded: DJI SDK, ESP32, Raspberry Pi, ROS/Gazebo, ArduPilot, MQTT, I2C, Fusion 360

Spoken: English (Fluent), German (B2 (CEFR))